COMP233 — Discrete Mathematics

Lecture 4.3 — Divisibility

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Why Divisibility Matters

- Divisibility is a relationship between integers—not merely a calculation.
- It supports precise mathematical proofs and algorithmic reasoning.
- Prime factors help explain properties of numbers without large calculations.
- Later: quotient–remainder, modular arithmetic, and computing applications.

Learning Goals

- Translate “a divides b” into an equation with an integer witness.
- Classify valid, invalid, and boundary cases of divisibility.
- Prove algebraic divisibility claims and transitivity directly.
- Refute false statements with a counterexample.
- Apply prime divisors and unique factorization to solve problems.

Definition: Divisibility

For integers a and b, with $a \neq 0$:

$$a \mid b \iff b = ak \text{ for some integer } k$$

a is a divisor (factor) of b; b is a multiple of a.

An integer witness k establishes the divisibility claim.

Words, Symbols, and Integer Witnesses

The relationship

“3 divides 21”

$3 \mid 21$

$21 = 3 \times 7$

Witness: $k = 7$

The distinction

$a \mid b$ is a statement (true or false).

a / b is a number (when $b \neq 0$).

For $a \neq 0$: $a \mid b$ iff b/a is an integer.

Quick Classification: Divides or Does Not Divide?

True examples

$3 \mid 21$ because $21 = 3 \times 7$

$5 \mid 40$ because $40 = 5 \times 8$

$7 \mid 42$ because $42 = 7 \times 6$

Negative examples

$-16 \mid 32$ because $32 = (-16)(-2)$

$7 \mid -7$ because $-7 = 7(-1)$

$-3 \mid 66$ because $66 = (-3)(-22)$

Boundary Cases: Zero

Zero as a multiple

For every nonzero integer a:

$a \mid 0$ because $0 = a \times 0$.

Example: $5 \mid 0$.

Zero as a divisor

Under the convention $a \neq 0$, zero is not an allowed divisor.

Do not write “$0 \mid b$” as a permitted divisibility claim.

A condition on the divisor matters.

Positive Divisors: An Important Bound

If $a,b > 0$ and $a \mid b$, then $a \leq b$.

- Proof: $b = ak$ for some integer k.
- Since $a,b > 0$, the integer k must satisfy $k \geq 1$.
- Multiply by $a > 0$: $b = ak \geq a$.
- For $b \neq 0$ and $a > 0$, the general bound is $a \leq |b|$.

Proving Divisibility: The Core Strategy

- State what must be proved: $d \mid E$.
- Rewrite the target expression as $E = d \times k$.
- Verify that k is an integer using integer closure.
- Conclude $d \mid E$ by the definition of divisibility.

The proof is not complete until the multiplier is known to be an integer.

Worked Example: 3 Divides 3a + 3b

Claim: For all integers a,b, 3 | (3a + 3b).

$$3a + 3b = 3(a + b)$$

Because a+b is an integer, it is a valid multiplier.

Therefore, 3 divides 3a + 3b.

Worked Example: Another Algebraic Expression

Claim: For all integers k,m, 5 | (10km).

$$10km = 5(2km)$$

The product 2km is an integer.

Therefore, 5 | 10km.

When Divisibility Fails

Non-example

Does 4 divide 15?

15/4 = 3.75 is not an integer.

Therefore $4 \nmid 15$.

Contrast

Does 4 divide 16?

$16 = 4 \times 4$, with integer multiplier 4.

Therefore $4 \mid 16$.

Prime Numbers and Divisibility

An integer $p > 1$ is prime iff its only positive divisors are 1 and p.

- Examples: 2, 3, 5, 7, 11 are prime.
- 6 is composite because $6 = 2 \times 3$.
- 1 is neither prime nor composite.
- Negative integers are not called prime under this definition.

Transitivity of Divisibility

If a | b and b | c, then a | c.

- Start: b = ar and c = bs for integers r,s.
- Substitute: c = bs = (ar)s = a(rs).
- Because rs is an integer, a | c.

How to Discover the Transitivity Proof

Given

$a \mid b$ means $b = a \times$ (integer).

$b \mid c$ means $c = b \times$ (integer).

Goal: express c as a times an integer.

Connect the equations

$b = ar$

$c = bs$

$c = (ar)s = a(rs)$

Witness for $a \mid c$: $k = rs$.

Prime Divisor Theorem

Every integer $n > 1$ has at least one prime divisor.

- If n is prime, then the prime divisor is n itself.
- If n is composite, write $n = ab$ with $1 < a < n$.
- Factor a further until a prime factor p appears.
- By transitivity, $p \mid a$ and $a \mid n$ imply $p \mid n$.

Why a Prime Divisor Must Exist

- Assume a nonempty set S of integers > 1 with no prime divisor.
- By well-ordering, choose the least member m of S.
- m cannot be prime, so m = ab with 1 < a < m.
- Minimality: a has a prime divisor p, hence p | m.
- Contradiction: m does have a prime divisor.

This is the minimal-counterexample / well-ordering proof idea.

Counterexample: A False Generalization

Claim: If $a \mid b$ and $b \mid a$, then $a = b$.

- Take $a = 2$ and $b = -2$.
- $2 \mid -2$ because $-2 = 2(-1)$.
- $-2 \mid 2$ because $2 = (-2)(-1)$.
- But $2 \neq -2$: therefore the universal claim is false.

Repairing the False Claim

Correct conclusion

For nonzero integers a,b:

If $a \mid b$ and $b \mid a$, then $|a| = |b|$.

Equivalently, $a = \pm b$.

Positive-integer special case

If $a,b > 0$ and $a \mid b$ and $b \mid a$, then $a = b$.

Because $a \leq b$ and $b \leq a$.

The Fundamental Theorem of Arithmetic

Every integer greater than 1 is prime or can be written as a product of prime numbers, uniquely up to the order of factors.

Example: $72 = 2 \times 2 \times 2 \times 3 \times 3 = 2^3 \times 3^2$

Reordering prime factors does not change the factorization.

Standard Prime-Factored Form

Write 3300 in standard prime-factored form.

$$3300 = 33 \times 100$$

$$= (3 \times 11)(4 \times 25)$$

$$= (3 \times 11)(2^2 \times 5^2)$$

$$\mathbf{= 2^2 \times 3 \times 5^2 \times 11}$$

Application: Must 17 Divide m?

$$8\times7\times6\times5\times4\times3\times2\times m = 17\times16\times15\times14\times13\times12\times11\times10$$

- The right-hand side contains the prime factor 17.
- None of the explicit left-hand factors 2,3,4,5,6,7,8 contains a factor 17.
- Unique prime factorization forces the factor 17 to occur in m.
- Conclusion: $17 \mid m$.

Concept Check: Test the Claim

Evaluate

A. Does $7 \mid 0$?

B. Does $0 \mid 7$? (standard convention)

C. Does $-6 \mid 42$?

D. Does $4 \mid 15$?

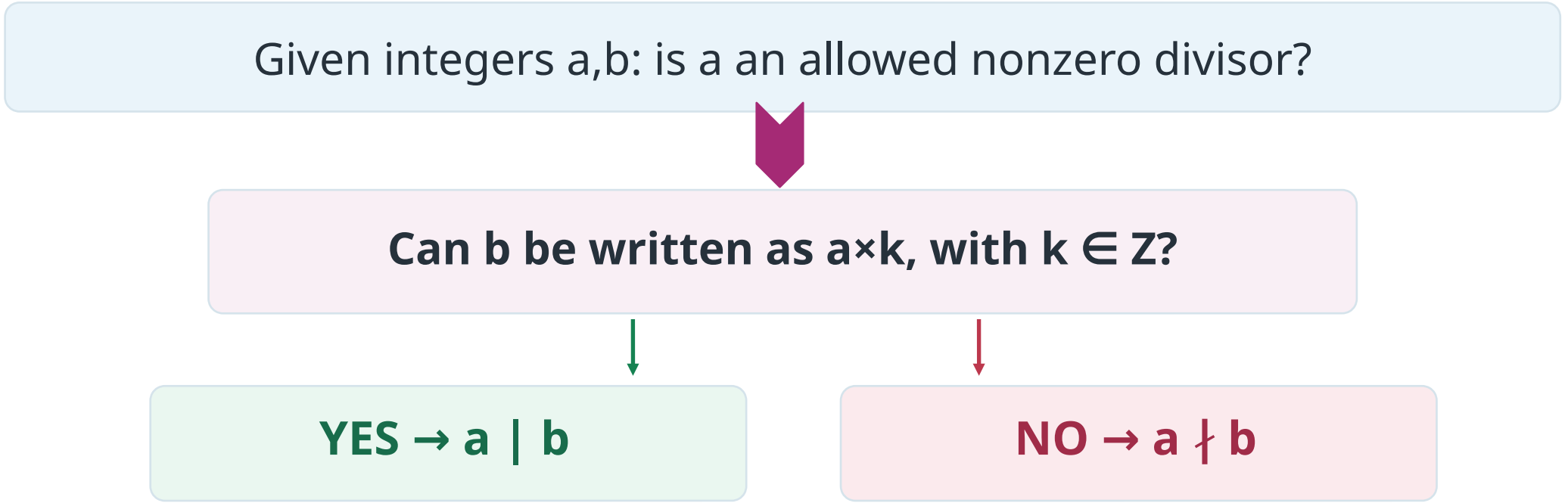
Justify

For each claim:

Find an integer witness, or show why no integer witness exists.

Check zero and sign conventions.

A Visual Decision Procedure



If a = 0: excluded under this lecture’s nonzero-divisor convention.

Appendix A — Core Concepts

- $a \mid b \Leftrightarrow b = ak$ for some $k \in \mathbb{Z}$, with $a \neq 0$.
- For every $a \neq 0$, $a \mid 0$; a divisor may be negative.
- Direct proof: construct an integer multiplier.
- Transitivity: $a \mid b$ and $b \mid c$ imply $a \mid c$.
- Every $n > 1$ has a prime divisor and unique prime factorization.

Appendix B — Conceptual Connections

- Definition → algebraic factorization → existence of integer witness.
- Mutual divisibility → equal absolute values (nonzero integers).
- Positive mutual divisibility → equality.
- Transitivity → transfer of prime factors across a product.
- Unique factorization → reasoning about divisibility without multiplying.

Appendix C — Contrasting Cases

Valid cases

$6 \mid 54$ because $54 = 6 \times 9$.

$7 \mid -7$ because $-7 = 7(-1)$.

$5 \mid 0$ because $0 = 5 \times 0$.

Invalid or special cases

$4 \nmid 15$ because $15/4 \notin \mathbb{Z}$.

0 is not an allowed divisor.

Mutual divisibility does not force $a=b$ for signed integers.

Appendix D — Knowledge Check 1/8

Which proves $5 \mid (15x-10)$ for all integers x?

A. $15x-10$ is positive. B. $15x-10=5(3x-2)$. C. $5<15$.

Explain your reasoning before checking the answer key.

Appendix D — Knowledge Check 2/8

True or false? Every nonzero integer divides 0.

Provide an integer witness.

Explain your reasoning before checking the answer key.

Appendix D — Knowledge Check 3/8

True or false? If $a \mid b$, then $b \mid a$.

Prove or give a counterexample.

Explain your reasoning before checking the answer key.

Appendix D — Knowledge Check 4/8

Find the error: 6 | (6a+3) because both terms are multiples of 3.

Explain what divisor is actually supported.

Explain your reasoning before checking the answer key.

Appendix D — Knowledge Check 5/8

If $a \mid b$ and $b \mid c$, show $a \mid c$.

Use integer multipliers.

Explain your reasoning before checking the answer key.

Appendix D — Knowledge Check 6/8

Can a prime integer greater than 1 have two distinct positive divisors?

Explain the meaning of “distinct”.

Explain your reasoning before checking the answer key.

Appendix D — Knowledge Check 7/8

Write 360 in standard prime-factored form.

Use prime powers.

Explain your reasoning before checking the answer key.

Appendix D — Knowledge Check 8/8

If p is prime, p | xy does p | x or p | y follow?

State your reasoning (Euclid’s lemma).

Explain your reasoning before checking the answer key.

Appendix E — Answer Key 1/8

Question 1: B

The witness is $k=3x-2 \in Z$. Options A and C do not establish an integer multiple.

Appendix E — Answer Key 2/8

Question 2: True

$0 = a \times 0$ for every nonzero integer a. This does not permit zero as a divisor.

Appendix E — Answer Key 3/8

Question 3: False

$2 \mid 6$, but $6 \nmid 2$. Divisibility does not generally reverse.

Appendix E — Answer Key 4/8

Question 4: The conclusion is unsupported

$6a+3=3(2a+1)$, so 3 divides it. Taking $a=0$ gives 3, which 6 does not divide.

Appendix E — Answer Key 5/8

Question 5: Direct proof

Write $b=ar$, $c=bs$. Then $c=a(rs)$, and $rs \in Z$.

Appendix E — Answer Key 6/8

Question 6: Yes—exactly two

For prime p, positive divisors are precisely 1 and p; they are distinct because p>1.

Appendix E — Answer Key 7/8

Question 7: $360 = 2^3 \times 3^2 \times 5$

$360=36\times10=(2^2\times3^2)(2\times5)$.

Appendix E — Answer Key 8/8

Question 8: Yes

Euclid’s lemma: for prime p, p | xy implies p | x or p | y. This is an enrichment beyond the original main lecture.

Appendix F — Applications and Enrichment

- Computer science: divisibility checks appear in loops and integer algorithms.
- Cybersecurity: primes and factorization underpin important number-theoretic constructions.
- Euclid’s lemma: for prime p, $p \mid ab$ implies $p \mid a$ or $p \mid b$.
- Next lecture: $n = dq + r$, where $d > 0$ and $0 \leq r < d$.

References and Course Resources

Primary reference

Susanna S. Epp, Discrete Mathematics with Applications, 4th ed., §4.3.

Original course materials

Prof. Mustafa Jarrar · https://www.jarrar.info/courses/DMath/

Updated course resources

https://github.com/RuwaYafa/BZU_Courses/tree/main/COMP233

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